



Instituto Nacional de Saúde
Doutor Ricardo Jorge



Fundação
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e a Tecnologia

Coupling Environmental Exposure and Mobility in Epidemiological Models

Application to Influenza Dynamics

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Respiratory infections such as influenza and RSV remain a **major public health burden** and exhibit **strong seasonal patterns** [1].

Three mechanisms have been proposed to explain **virus seasonality** – **Host Resistance**, **Virus Survival** [2, 3] and **Behavioural Changes** [4].

Rather than acting as direct causes, **environmental factors act as modulators of transmission intensity** [5].

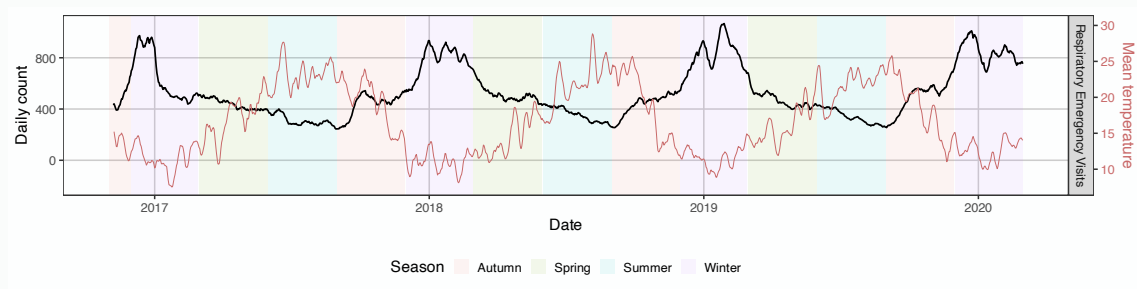


Figure 1: 7-day moving averages of respiratory emergency visits in Health Administrative Region Lisboa and Tagus Valley. Red line shows 7-day moving averages mean temperature. Coloured backgrounds indicate season.

Objective

Research Question

How does coupling temperature-dependent transmission with human mobility alter influenza dynamics in a SEIR framework?

Objectives

1. Implement a stochastic SEIR-ABM with **temperature-dependent transmission** and **human mobility** coupling
2. Derive an equivalent deterministic ODE system under the same coupling
3. Simulate influenza dynamics using **temperature** and **mobility** data from Portugal
4. Compare the introduction of mobility and temperature on system outcomes

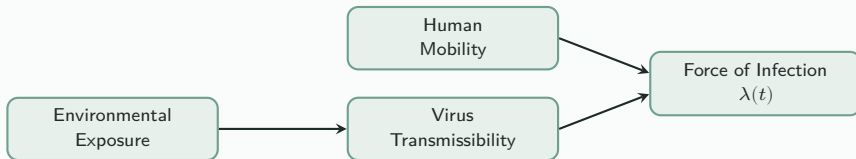


Figure 2: Impact of environmental conditions and mobility on transmission.

We consider a SEIR-type metapopulation model with P patches,

Within-patch dynamics ($i = 1, \dots, P$)

- Infection: $S_i \rightarrow E_i$ at rate $\lambda_i(t)$
- Progression: $E_i \rightarrow I_i$ at rate σ
- Recovery: $I_i \rightarrow R_i$ at rate γ

$$S'_i = -\lambda_i(t) S_i$$

$$E'_i = \lambda_i(t) S_i - \sigma E_i$$

$$I'_i = \sigma E_i - \gamma I_i$$

$$R'_i = \gamma I_i$$

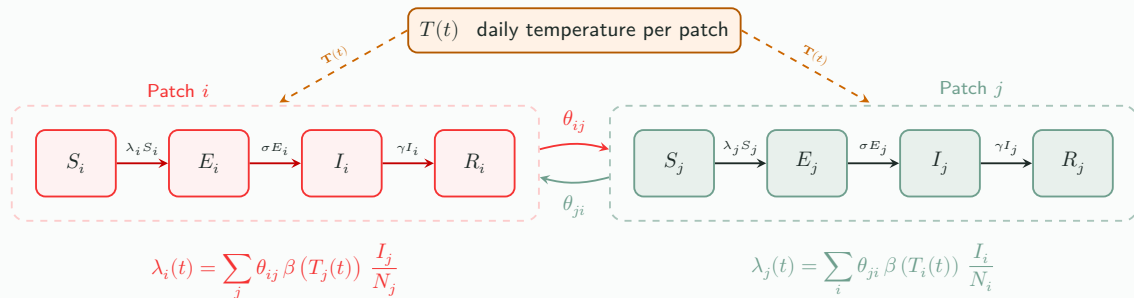


Figure 3: Metapopulation SEIR with temperature-dependent transmission and mobility coupling $\theta = [\theta_{ij}]$

Figure 4: Daily origin–destination pendular mobility across LVT municipalities.

Inter-municipal mobility is derived from an Origin–Destination matrix of **pendular movements**, normalized to represent the fraction of time residents of patch i spend in patch j .

OD data was taken from the **Portuguese Census 2021** [6].

Temperature Coupling

Patch-specific temperature time series $T_j(t)$ drive local transmissibility and contact rates through $\beta(T_j(t))$, all parameters equal across all P patches.

A **logistic sigmoid** maps daily temperature to a transmission scaling factor κ :

$$\kappa(T) = y_{\min} + \frac{\Delta y}{1 + e^{k_T(T - T_{\text{mid}})}}$$

$$\beta(T_j(t)) = \kappa(T_j(t)) \cdot \beta$$

Daily gridded observational temperature dataset for Europe (E-OBS) from the **Copernicus Climate Data Store** [7].

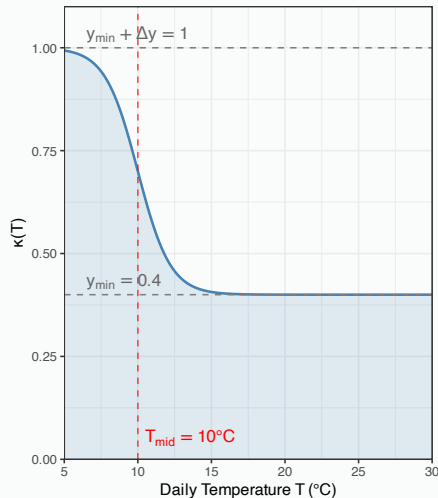


Figure 5: Sigmoid $\kappa(T)$ mapping daily temperature to a transmission scaling factor. Red dashed line marks $T_{\text{mid}} = 10^\circ\text{C}$.

Use Case: Influenza H3N2 season with $P = 7$ spatial patches.

Data: Patch temperatures $T_i(t)$ from E-OBS (Copernicus); mobility matrix θ from OD pendular flows, Census 2021 – *borrowed* from specific municipalities in Lisboa e Vale do Tejo.

Parameters: Latent period $1/\sigma = 1.9$ days, infectious period $1/\gamma = 4.1$ days, $\beta = 0.42 \text{ d}^{-1}$, $T_{\text{mid}} = 10^\circ\text{C}$, yielding $R_0 \approx 2.0$.

Agents: Total of 11 200 agents with 15 seeded infectious

Scenarios: Design toggles mobility (θ vs. $\mathbf{I}_P$) and temperature ($\kappa(T)$ vs. $\kappa \equiv 1$) independently.

	Scenario	Mobility	Temp.	Purpose
S1	Baseline	$\mathbf{I}_P$	$\kappa \equiv 1$	Homogeneous reference
S2	Temperature only	$\mathbf{I}_P$	$\kappa(T)$	Seasonal forcing alone
S3	Mobility only	θ	$\kappa \equiv 1$	Spatial spread alone
S4	Full model	θ	$\kappa(T)$	Combined mechanisms

Simulation Results

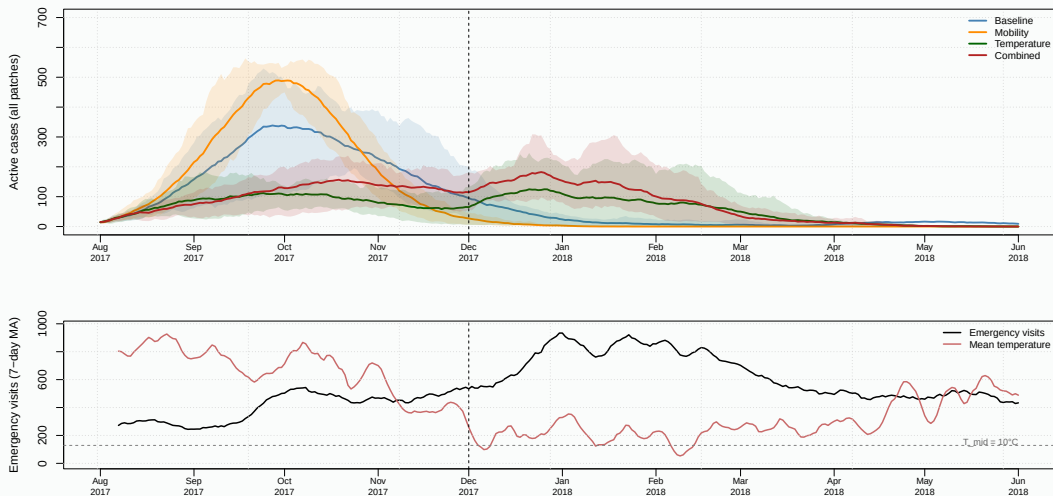


Figure 6: Simulated influenza dynamics compared to observed epidemiological and environmental data for the 2017–18 season. (*Top*) Global daily active cases (mean $\pm$ 10–90% interquartile range across 15 replicates) under four scenarios: Baseline (no mobility, no temperature coupling), Mobility (commuting matrix active), Temperature (sigmoid scaling of β by local temperature), and Combined (both mechanisms active). (*Bottom*) Observed respiratory emergency visits in ARS Lisboa e Vale do Tejo and mean daily temperature.

Discussion

Constant Parameters (Null Model)

- **Driver** Epidemic trajectory determined solely by R_0 and initial seeding.
- **Outcome** Peak occurs too early or is overly broad, driven purely by the depletion of susceptibles.

Combined Model (Temperature + Mobility)

- **Driver** $\beta(T(t))$ moderated by environmental exposure; cross-patch FOI shaped by θ .
- **Outcome** Growth suppressed during warm months, delaying the exponential phase into Winter.

Coupling **temperature-dependent transmission** with **human mobility** produces an epidemic curve whose **timing, shape, and winter peak** are qualitatively consistent with the **hospitalisation and emergency visit patterns** observed.

Key Takeaways

- **Environmental drivers matter** the sigmoid $\kappa(T)$ alone shifts the peak into the epidemiologically correct winter window; mobility broadens and redistributes it spatially.
- **Mechanistic flexibility** time-varying $\beta(T(t))$ can absorb **non-pharmaceutical interventions** independently of environmental forcing; the mobility matrix θ can be swapped for real-time data.
- **Multi-strain extensibility**
- **Data dependence** the framework requires external **temperature** and **mobility** data to inform transmission — a strength for calibration, but a limitation in data-scarce settings.

Thank you for your attention!

Any questions?

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Contacts

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github.com/andrebrito0



Scan for the GitHub repository with slides, data
and references.

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Additional Slides

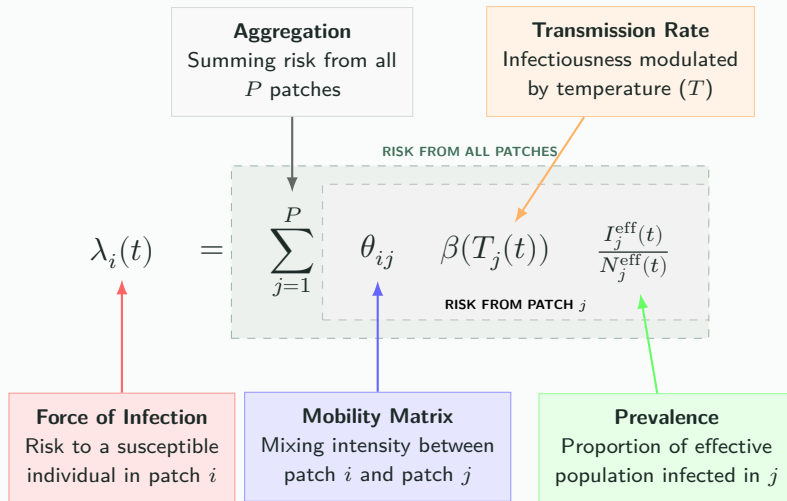


Figure 7: Schematic decomposition of the force of infection, $\lambda_i(t)$. The equation aggregates the risk across all patches visited, weighted by the mobility matrix θ and the temperature-dependent transmission rate β_j .

disease	
name	chr
beta_core	β household
beta_ext	β community
sigma	$1/t_{\text{incub}}$
gamma	$1/t_{\text{infect}}$

agent	
id	int
age	int
status	S/E/I/R
timer	time-to-transition
core_contacts	HH agent IDs
ext_contacts	λ daily contacts

population	
agents	list(agent)
disease	disease obj
k_core_pop	mean HH size
k_ext_pop	mean k_{ext}
history	SEIR counts/day

disease_spatial	
$\hookrightarrow$ inherits disease	
y_min	floor κ
delta_y	range κ
k_temp	steepness
T_mid	inflection ($^{\circ}\text{C}$)

patch	
id, name	int, chr
pop_size	int
S, E, I, R	aggregate counts
history	data.frame

metapopulation	
patches	list(patch)
agents	flat agent pool
patch_ids	home patch map
disease	disease_spatial
Phi	mobility matrix Φ
temp_series	$T_i(t)$ matrix

methods	
update_agent	$E \rightarrow I, I \rightarrow R$
assign_core_group	Poisson HH
assign_infected	seed I_0
step_sim	single-patch FOI
step_metapop	spatial FOI + $\kappa(T)$
summary / plot_*	diagnostics

- Blue — identifiers & transmission params.
- Teal — cross-object references.
- Orange — derived/modelled quantities.

- disease_spatial adds sigmoid $\kappa(T)$ for spatial FOI.
- Phi is row-stochastic: $\sum_j \Phi_{ij} = 1$ per origin patch.
- Generics dispatch on first-argument class (S7).

Temperature Data

Daily gridded observational dataset for Europe (E-OBS) from the **Copernicus Climate Data Store** [7].

Mobility Matrix

Pendular movement flows from the **2021 Portuguese Census** (INE).

Row-normalised matrix: Φ_{ij} gives the proportion of time residents of patch i spend in patch j ,

$$\sum_{j=1}^P \Phi_{ij} = 1 \quad \forall i$$

Population Initialisation

- P spatial patches, each with N_i agents drawn from census population sizes
- Each agent assigned to a **core group** (household) via Poisson draw: $k_{\text{core}} \sim \text{Pois}(\bar{k}_{\text{hh}}) + 1$
- External sociality drawn from a Gamma distribution: $k_{\text{ext}} \sim \text{Gamma}(2, 2/\bar{k}_{\text{ext}})$
- I_0 agents seeded as infectious in selected patches at $t = 0$

Temperature Coupling $\kappa(T)$

A **logistic sigmoid** maps daily temperature to a transmission scaling factor $\kappa \in [y_{\min}, y_{\min} + \Delta y]$:

$$\kappa(T) = y_{\min} + \frac{\Delta y}{1 + e^{k_T(T - T_{\text{mid}})}}$$

The effective transmission rate becomes:

$$\beta_i(t) = \kappa(T_i(t)) \cdot \beta_{\max}$$

Cold conditions ($T \ll T_{\text{mid}}$) push $\kappa \rightarrow 1$; warm conditions suppress transmission toward $y_{\min} = 0.20$.

Scenario comparison: four runs toggle mobility (vs. $\mathbf{I}_P$) and temperature ($\kappa(T)$ vs. $\kappa \equiv 1$) independently, isolating each mechanism's contribution to outbreak dynamics.